

THARUN ARETY

Agentic Information Engineer | AI Agents, RAG & MCP for Engineering Data

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PROFILE

AI systems builder with an engineering background. I build agentic ERP, CRM and PDM systems, RAG pipelines and MCP integrations that turn fragmented business data into something a team can query. I own the full stack myself: Python/TypeScript services, Postgres/pgvector infrastructure, and the eval suites and guardrail pipelines that keep agents safe to run.

EXPERIENCE

AI Systems Architect / AI Engineer 2025 – 2026

- Vexos Remote
- Replaced costly ERP, CRM, and PDM software by architecting a custom end-to-end agentic toolchain, eliminating SaaS dependencies.
 - Slashed compliance processing time by 96% (60 mins to <2 mins per batch) by engineering a vision-based AI pipeline with automated schema validation.
 - Unlocked natural-language querying for siloed business data by deploying RAG pipelines, MCP servers, and tool-calling agents for executable AI workflows.

Research Associate, 2023 – 2026

Data-driven Product Engineering & Design (MRM) Augsburg, DE

- Scaled computational engineering throughput by fully automating the simulation lifecycle across expansive design spaces.
- Accelerated design space evaluations by building PyTorch/JAX surrogate models and differentiable optimization frameworks to bypass slow traditional methods.
- Accelerated software development and technical automation cycles by systematically integrating LLMs and coding agents into daily research workflows.

PROJECTS

Magnetocaloric Engineering Agent

- Next.js 16 • OpenAI SDK (no framework) • PostgreSQL + pgvector (Neon) • ajv
- Achieved 95.8% reliable data retrieval (138/144 test cases) by designing a dual-agent system secured by a three-layer guardrail architecture.

Agentic PDM / ECM / QMS Toolchain

- FastAPI • LangGraph • PostgreSQL + pgvector • SQLAlchemy 2 (async)
- Streamlined change-management by unifying quality data behind a LangGraph/ FastAPI interface, enabling AI-proposed changes with human-in-the-loop approvals.

Autonomous Compliance System:

- Eradicated manual compliance tracking via an autonomous vision system that manages certificate extraction, expiry monitoring, and supplier outreach.

TalentFlow

- Next.js/Postgres/Drizzle
- Optimized recruitment pipelines by deploying a custom Next.js/ Postgres ATS managing the entire lifecycle from requisition to hiring.

PERSONAL DETAILS

Nationality Indian

Work Status Authorised to work in Germany (No Sponsorship Required)

Availability Immediate

EDUCATION

M.Sc. Materials Engineering

University of Augsburg 2022 – 2026 Germany

B.Tech. Mechanical Engineering

NIT Agartala 2016 – 2020 India

SKILLS

Agentic & AI Systems

- AI Agents LLM APIs
- Tool Calling / Function Calling LangGraph
- RAG Vector / Hybrid Retrieval Embeddings
- MCP Servers Prompt Engineering Coding Agents

AI Reliability & Guardrails

- Evals LLM-as-Judge Guardrail Pipelines
- Schema Validation (ajv) Tool-Argument Validation
- Human-in-the-Loop Observability

Software & Infrastructure

- Python TypeScript FastAPI
- Next.js React Node.js
- PostgreSQL pgvector SQLAlchemy (async)
- Docker Linux REST APIs Git

Enterprise & PDM Systems

- ERP CRM PDM ECM QMS
- BOM & Engineering-Change Mgmt
- Compliance Automation Knowledge Mgmt
- Document Intelligence

LANGUAGES

English (C1) German (B1 → C1 in progress)

ACHIEVEMENTS

Scholarship Holder– Albert Leimer Stiftung & DAAD
Awarded for academic excellence in M.Sc. Materials Engineering at the University of Augsburg.